

Question 1:

a. **2** pairs (2 parameters for translation, and one for rotation).

b. **1** pair (1 parameter for scale).

c. **4** pairs (8 parameters for homography).

d. **3** frames :

$$TTC = (x_2 / (x_2 - x_1)) - 1 = (4x_1/3 / (4x_1/3 - x_1)) - 1 = 4 - 1 = 3$$

e. **10** pixels:

$$Z_1 = \frac{f*b}{d} \Rightarrow 8 = \frac{f*b}{1} \Rightarrow f * b = 8$$

$$Z_2 = \frac{f*b}{d} = \frac{8}{0.8} = 10$$

f. Area **5** (the epipole is the projection of the center of the second camera on the first image $\Rightarrow$ the center of the camera is the same in both images $\Rightarrow$ epipole in the center).

Question 2:

The problem can be broken down into 4 parts:

1. Choosing reference frame (1 point)
2. Alignment - finding the translation between images (3 points)
3. Warping (2 points)
4. Stitching (2 points)
 - The correct algorithm for alignment is **Lucas-Kanade** (direct alignment). Feature based alignment will not work, as there are no pixels which either MOPS or SIFT will detect as feature points. Using NCC will be much less efficient.
 - The correct algorithm for warping is **backward warping** (does not suffer from 'holes' in the output).
 - The correct algorithm for stitching is **min-cut or dynamic programming** (in order for the seam to go around the stars, if they exist).